

Deep Unfolding and Failure Modes in Joint Sparse Recovery with Unknown Illumination

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Abstract

We study a block-coordinate optimizer for compressive imaging when the measured signal is the product of a sparse scene and an unknown smooth illumination field. The coefficient block is a nonsmooth Lasso problem solved by FISTA, while the illumination block is a regularized least-squares problem. On five natural-image patches, joint optimization improves PSNR by up to 4.7 dB on textured scenes but loses 7.8–10.0 dB on a low-energy scene. A natural conditioning-number gate fails: the worst scene has the *best* measured conditioning. Separately, over five optimizer seeds a ten-layer tied LISTA reaches NMSE 0.0435 ± 0.0120 , which FISTA needs 23 iterations to match. This $2.3\times$ iteration reduction does not yield a batch-1 CPU latency gain at $N = 200$ (0.664 versus 0.484 ms). We characterize both the scene-energy failure and the poor scaling of the present dense implementation. The results show that identifiability and operator structure, not iteration count alone, govern a practical joint optimizer.

1. Introduction

Optimization-based imaging often separates reconstruction from image-signal processing. This ordering is questionable when a nuisance field changes the very representation on which the reconstruction prior relies. We consider measurements

$$y = A \operatorname{diag}(g_{\text{pix}}) \Psi c + w, \quad (1)$$

where $c \in \mathbb{R}^N$ is sparse, Ψ is a two-dimensional DCT synthesis operator, $g \in \mathbb{R}^q$ is a smooth per-column illumination vector with $q = \sqrt{N}$, and g_{pix} tiles g over image rows. Recovering the illuminated image and correcting it afterwards applies the sparsity prior to the product rather than the scene. We instead optimize both variables. This places the study in classical compressed sensing [2, 3], with an additional bilinear nuisance variable.

This is a small-scale, deliberately diagnostic study. Its contribution is not a claim of a production RAW pipeline. Rather, it asks three optimization questions: (i) when does block-coordinate joint recovery help relative to a sequential optimizer, (ii) how much iteration reduction is available from a finite-depth learned proximal map, and (iii) what prevents the present method from scaling? Our main findings are:

- joint recovery is strongly scene dependent, with gains on textured scenes and a reproducible low-energy failure;
- $\kappa(B^\top B)$ does not predict that failure, falsifying a natural deployment heuristic; and

- tied LISTA reduces the matched-quality iteration count by $2.3\times$, but does not improve measured batch-1 CPU latency at $N = 200$; dense operator formation and dense learned weights remain the dominant scaling barriers.

2. Joint block-coordinate optimization

We minimize the block-nonconvex objective

$$\min_{c, g \geq \epsilon_g} \frac{1}{2} \|A \text{diag}(g_{\text{pix}}) \Psi c - y\|_2^2 + \lambda_c \|c\|_1 + \lambda_g \|Dg\|_2^2, \quad (2)$$

where D is a first-difference operator. With g fixed, define $A_g = A \text{diag}(g_{\text{pix}}) \Psi$ and apply T_c FISTA steps to

$$\min_c \frac{1}{2} \|A_g c - y\|_2^2 + \lambda_c \|c\|_1 \quad [1]. \quad (3)$$

With c fixed, let $s = \Psi c$ and construct $B \in \mathbb{R}^{M \times q}$ so that $Bg = A \text{diag}(g_{\text{pix}}) s$. The gain update is

$$g \leftarrow (B^\top B + \lambda_g D^\top D + \eta I)^{-1} B^\top y, \quad g \leftarrow \max(g, \epsilon_g). \quad (4)$$

We use eight outer iterations. The model has the unavoidable ambiguity $(s, g) \mapsto (as, g/a)$; both pipelines therefore receive the same scalar gain-matching operation before evaluation.

What is and is not unfolded. For the coefficient problem, ISTA has the recurrence

$$c^{k+1} = \mathcal{S}_{\theta_k}(W_t c^k + W_e y). \quad (5)$$

Tied LISTA learns W_e , W_t , and per-layer thresholds for $K = 10$ layers, initialized from the ISTA operator [4]. Our LISTA experiment tests this coefficient solver on a fixed synthetic sparse-recovery problem. The reported joint-image experiments still use FISTA in the coefficient block. Thus the evidence supports LISTA as a candidate replacement for the inner loop; it does *not* yet establish an end-to-end unfolded outer optimizer.

Relation to prior unfolding. LISTA introduced the learned finite-depth sparse coder [4]; ISTA-Net adapted optimization-inspired unrolling to image compressive sensing [6]. Broader unrolling work emphasizes that interpretability and speed depend on preserving useful algorithmic structure [5]. Our narrower contribution is the diagnostic separation between inner-solver acceleration and failure of the surrounding bilinear optimization.

3. Scaling analysis

The present NumPy implementation explicitly stores $A \in \mathbb{R}^{M \times N}$ and a dense $\Psi \in \mathbb{R}^{N \times N}$. It also materializes A_g at every outer step, costing $O(MN^2)$ time. A tied LISTA layer costs $O(N^2 + MN)$ and stores $O(N^2 + MN)$ values, whereas a FISTA step costs $O(MN)$. Table 1 measures the coefficient solvers directly.

If $M = \delta N$, explicit formation is cubic in N and tied LISTA has quadratic parameters; neither is suitable for a megapixel-wide problem. Current image experiments avoid this by using 16×16 blocks, and the joint experiment uses only one center patch per scene. A scalable version must avoid A_g materialization, apply the separable DCT implicitly in $O(N \log N)$, replace dense sensing by

Table 1: Batch-1 float32 CPU benchmark (one thread): 10 LISTA layers versus 23 FISTA iterations, with setup excluded. Memory is conservative working state in MiB.

N	LISTA ms	FISTA ms	LISTA params	memory L/F
64	0.470	0.542	5.8k	.030/.008
200	0.664	0.484	56.0k	.279/.066
256	0.763	0.482	91.7k	.455/.106
1024	8.650	3.942	1.47M	7.224/1.628

a structured operator, and use a factorized or convolutional unfolded map. Blockwise processing then scales linearly in the number of blocks, at the cost of blocking artifacts and a gain field that must be coupled across blocks. These are architectural requirements, not implementation micro-optimizations.

4. Experiments

All experiments use seeded RNGs. The image set comprises cameraman, astronaut, coins, page, and moon from `scikit-image`. Images are grayscale and rescaled to $[0, 1]$. Measurements use column-normalized Gaussian matrices. This is a controlled natural-image benchmark, not a sensor RAW dataset.

Inner-solver acceleration. LISTA is trained on 5000 synthetic pairs with $N = 200$, $M = 80$, sparsity $S = 10$, and SNR 30 dB; 500 pairs are held out. The sensing operator and dataset are fixed while optimizer initialization is repeated for five seeds on an RTX 8000 GPU. Table 2 compares equal depth and quality matched to the five-seed mean.

Table 2: Validation NMSE for the coefficient subproblem.

Solver	iterations/layers	NMSE
ISTA	10	0.4667
FISTA	10	0.3369
LISTA (5 seeds)	10	0.0435 ± 0.0120
FISTA (matched)	23	≈ 0.0435
FISTA (converged)	500	0.0050

Joint versus sequential recovery. We extract one 16×16 center patch per scene, apply an $8 \times$ horizontal illumination gradient, and add noise at SNR 25 dB. Sequential recovery applies 400 FISTA iterations followed by column-mean normalization. Joint recovery uses eight outer iterations and 120 FISTA steps per coefficient update. Table 3 reports joint-minus-sequential PSNR.

The mean over all five scenes favors sequential recovery because moon is an outlier. Excluding moon, joint recovery gains 0.4–1.5 dB across all measurement rates. We therefore make a conditional claim: joint estimation helps when the first coefficient step recovers enough scene energy to support gain estimation.

Table 3: Per-scene PSNR change (dB), joint minus sequential.

M/N	camera	astronaut	coins	page	moon
0.20	2.87	0.54	-1.02	-0.79	-10.03
0.30	4.65	0.18	-1.73	-0.48	-9.51
0.40	1.13	1.84	0.39	-0.23	-7.76
0.50	1.08	2.31	0.74	0.74	-9.05
0.60	3.07	2.21	0.32	0.43	-8.37

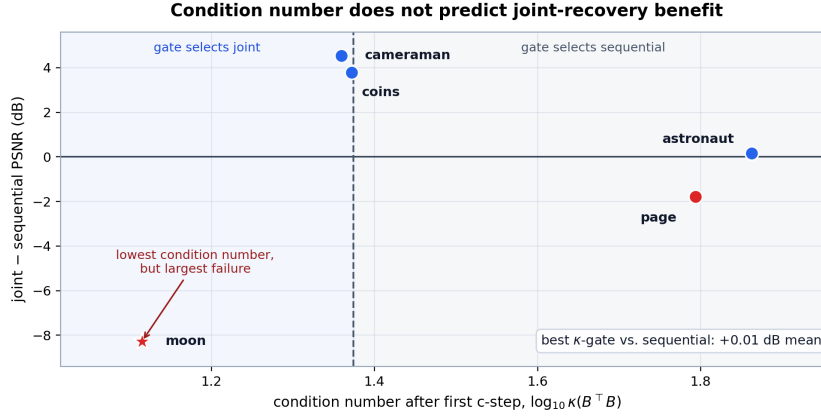


Figure 1: Per-scene joint-minus-sequential PSNR. The best threshold (dashed) still sends moon—the lowest- κ scene and largest loss—to joint.

Falsifying a natural gate. At $M/N = 0.4$, moon loses 8.27 dB even though it has the lowest observed $\log_{10} \kappa(B^T B) = 1.11$; cameraman gains 4.54 dB at a larger value of 1.36. The best threshold sweep improves mean PSNR by only 0.01 dB over sequential-only recovery (Figure 1). Conditioning measures relative sensitivity, but not the absolute scale of B : a low-energy scene can yield a well-conditioned matrix whose entries are too small relative to noise. A useful gate should instead measure a normalized scene-energy or signal-to-noise statistic after the first coefficient update. This gate is a hypothesis for future validation, not a completed result.

5. Discussion and limitations

The experiments expose three distinct limits. First, reducing inner iterations cannot repair a non-identifiable illumination update. Second, training stability is itself an optimization issue: the learned recurrence can diverge after its best checkpoint as the spectral radius of W_t drifts. Gradient clipping and best-checkpoint restoration contain the symptom, but a spectral constraint would be more principled. Third, the dense representation dominates asymptotic cost; a ten-layer network is not “fast” if every layer contains a full $N \times N$ map.

The evidence is intentionally limited: five images, synthetic illumination, Gaussian measurements, one joint patch per scene, five optimizer seeds on one fixed LISTA operator/dataset, and no modern learned reconstruction baseline. In particular, we do not call these measurements camera RAW data and do not claim end-to-end learned joint recovery. Before a camera-facing claim, the method needs

a real RAW benchmark, structured sensing, operator-distribution generalization, and comparison with a learned inverse-problem baseline.

6. Conclusion

Joint recovery helps textured scenes but fails when scene energy is too low, a regime conditioning does not detect. LISTA reduces inner iterations, but scaling requires structured implicit operators, constrained unfolding, and an energy-aware fallback.

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